



CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 02



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Recap of Previous Lecture



Topic

Basic of Graph



Topics to be Covered



Topic

Graph Part-01





Topic : Graphs



$$\sum d(v_i) = 2(e)$$

$$\sum d(v_i) = \text{even}$$

$$\underbrace{d(v_1)}_{\text{ODV}} + \underbrace{d(v_2)}_{\text{EDV}} + \underbrace{d(v_3)}_{\text{ODV}} + \dots + d(v_n) = \text{even}$$

$$\boxed{\overset{1}{0}, \overset{3}{0}, \overset{5}{0}, \dots} + \boxed{e, e, e, \dots} = \text{even}$$

✓ even = even

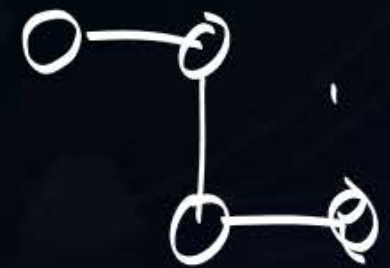
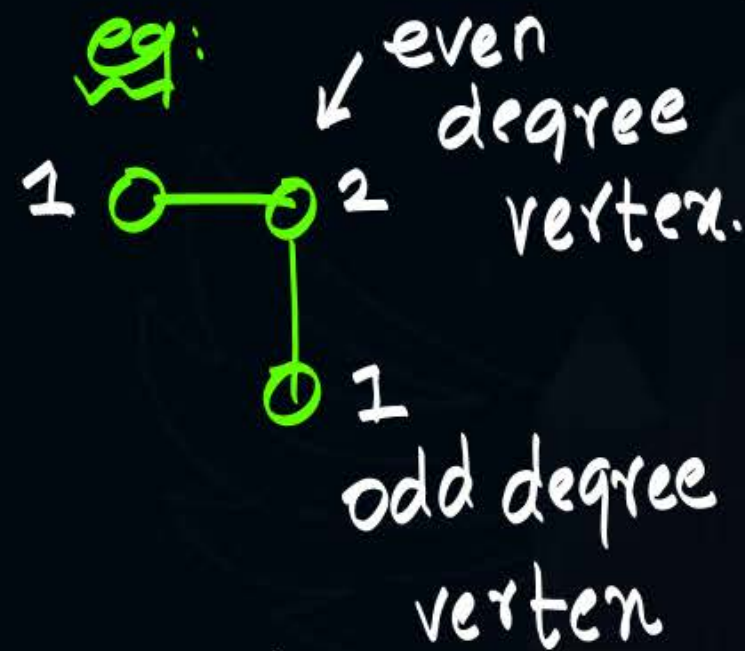
even. + even = even

odd

$$1+3 = \text{even}$$
$$1+3+5 = \text{odd}$$

1) $\sum d(v_i) = 17$
(false)

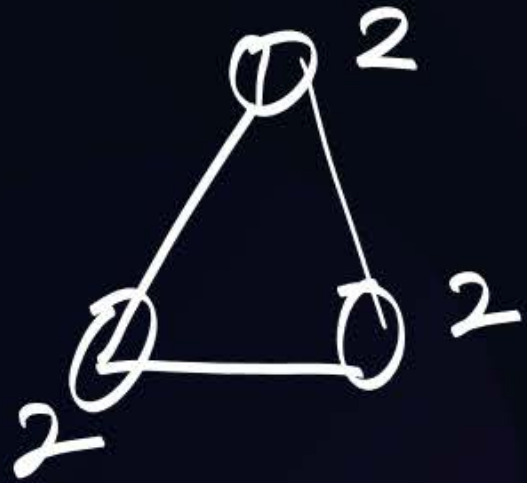
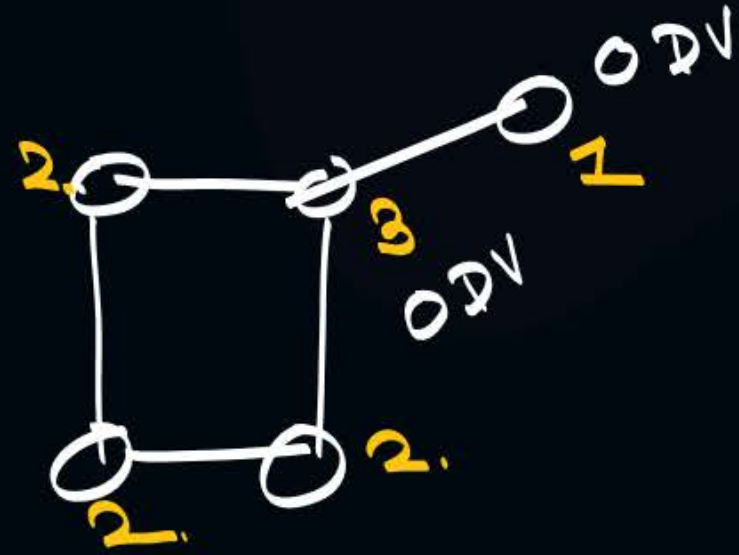
2) Graph can have
3 no. of edges.
(True)



$$\# \text{ of ODV} = 2$$

$$e + e = e$$

$$e + e + e = e.$$



$$\# \text{ of } ODV = 0.$$

$$\boxed{1+3}^{ODV} + \boxed{2+2+2}^{EDV} = \text{even}.$$



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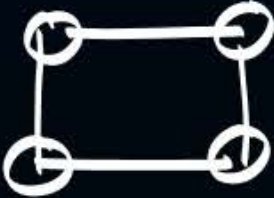
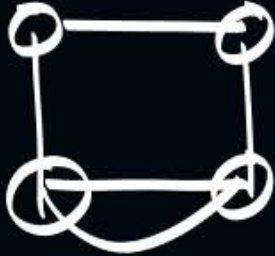
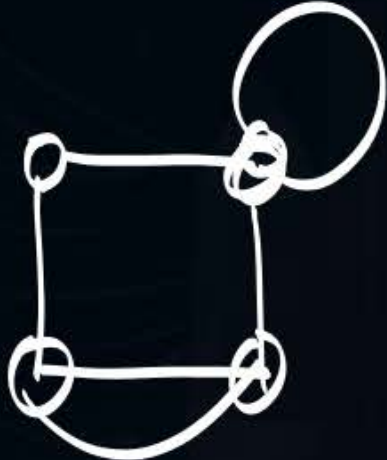


Thm 2 : no. of odd degree vertices in a graph will always be even.



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	loop	edges.	
Simple.	X	X.	
multigraph	X.	✓	
Pseudograph.	✓	✓	

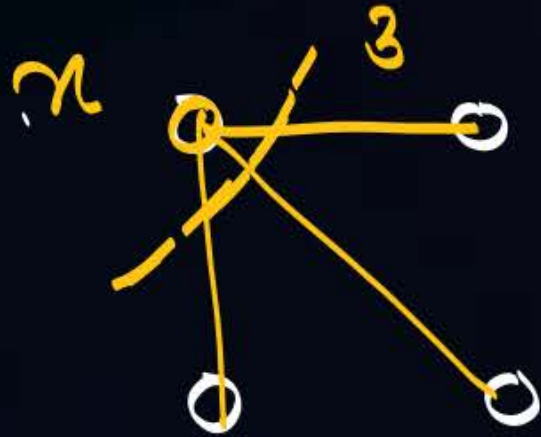


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Thm 3 : maximum degree in a simple Graph. $\leq n-1$.

Total vertices $n = 4$.



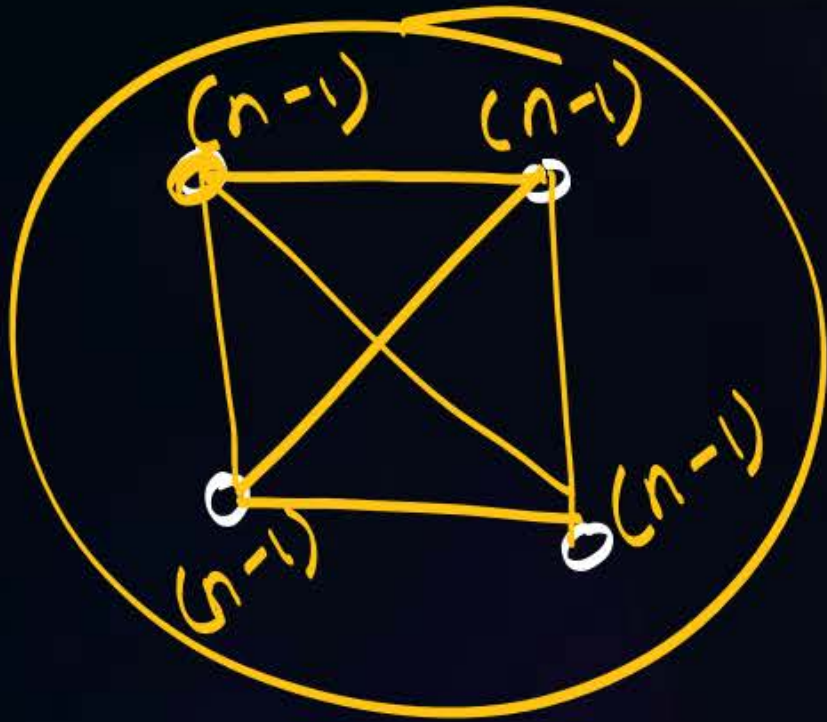


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Thm 4: maximum no. of edges in Simple Graph $\leq \frac{n(n-1)}{2}$.

$$n = 4.$$



Total vertices = n .

$$\sum d(v_i) = 2e.$$

$$n \times (n-1) = 2e.$$

$$e = \frac{n(n-1)}{2}.$$



Topic : Graphs



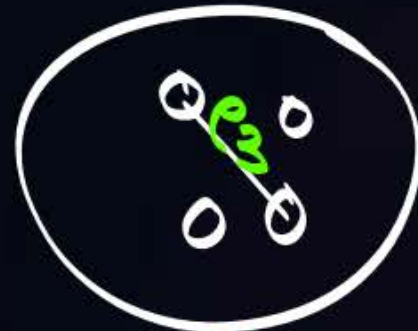
$n = 4$



$e = 0$

$n = 4$

$e = 1$



e_4

$n = 4$

$e = 2$



$n = 4$

$e = 3$

$n = 4$

$e = 4$

$n = 4$

$$e = \frac{n(n-1)}{2} = \frac{4 \cdot 3}{2} = 6$$

$n = 4$

$e = 6$

→ no simple
graph.

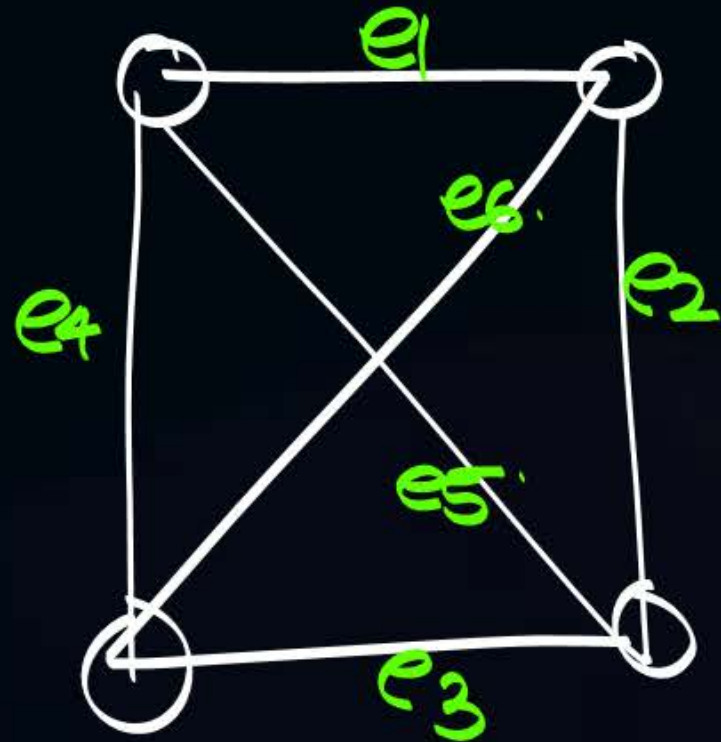
$$e = \frac{n(n-1)}{2}$$





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$n=4$ $e=6$



edge $\rightarrow$ present.
 $\rightarrow$ absent.



$6C1$

2^6
 $= 64$



2 2 2 2 2 2
 e_1 e_2 e_3 e_4 e_5 e_6

0	0	0	0	0	0
1	0	0	0	0	0
0	1	0	0	0	0

1 1 1 1 1 1



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note: if Graph is having n vertices.
Total possible graphs are

note: Total no. of graphs if we have.
 n vertices & ' e ' edges.

$$\rightarrow \frac{n(n-1)}{2} \subset e.$$

eg: $n=4$.

$$2^{\frac{n(n-1)}{2}}$$
$$\text{Total Graphs} = 2^{\frac{n(n-1)}{2}}$$
$$= 2^{\frac{4 \cdot 3}{2}}$$
$$= 2^6$$

$\Rightarrow$ what will be total no. of graphs
if we have 4 vertices & 1 edges.
Total 6 edges, we can choose
any 1 of the edges.
 $6C1$.



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what will be total no. of graphs with 4 vertices
at least 2 edges?

$$\text{Ans: } 6c_2 + 6c_3 + 6c_4 + 6c_5 + 6c_6.$$

Requires:

method 2: $6c_0 + 6c_1 + \boxed{6c_2 + 6c_3 + 6c_4 + 6c_5 + 6c_6} = 2^6.$

$$\text{Ans} = 2^6 - 6c_1 - 6c_0.$$



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What will be total graphs with 7 vertices & at least 3 edges.

at least 3 edges / 7 vertices.

$$n = 7. \text{ Total graphs} = \frac{n(n-1)}{2} = \frac{7 \cdot 6}{2} = 21.$$

$$2^{1c_0} + 2^{1c_1} + 2^{1c_2} + \dots + 2^{1c_{21}} = 2^{21}.$$

$$\text{Ans: } 2^{21} - 2^{1c_0} - 2^{1c_1} - 2^{1c_2}$$

b) at most 3 edges?

b) at most 3 edges.

$$\text{Ans: } = 2^{1c_0} + 2^{1c_1} + 2^{1c_2} + 2^{1c_3}$$



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GATE:-04.

Consider a Graph is having 27 edges.

3v of degree 4.

Remaining
vertices = x .

6v of degree 2.

remaining vertices of degree 3
what will be Total vertices?



$$\sum d(v_i) = 2e.$$

$$3 \times 4 + 6 \times 2 + x \cdot 3 = 2 \cdot 27$$

$$12 + 12 + 3x = 54.$$

$$3x = 30$$

$$x = 10$$

Ans: 19

$$\text{Total vertices} = x + 6v + 3v.$$

$$= 10 + 6 + 3 = \underline{19}$$



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Q: what will be max. no. of vertices if $e = 15$, degree of each vertex is at least 3?

Soltⁿ: $e = 15$ $\delta(G) = 3$

$$\delta(G) \leq \frac{2e}{n} \leq \Delta(G) \leq n-1$$

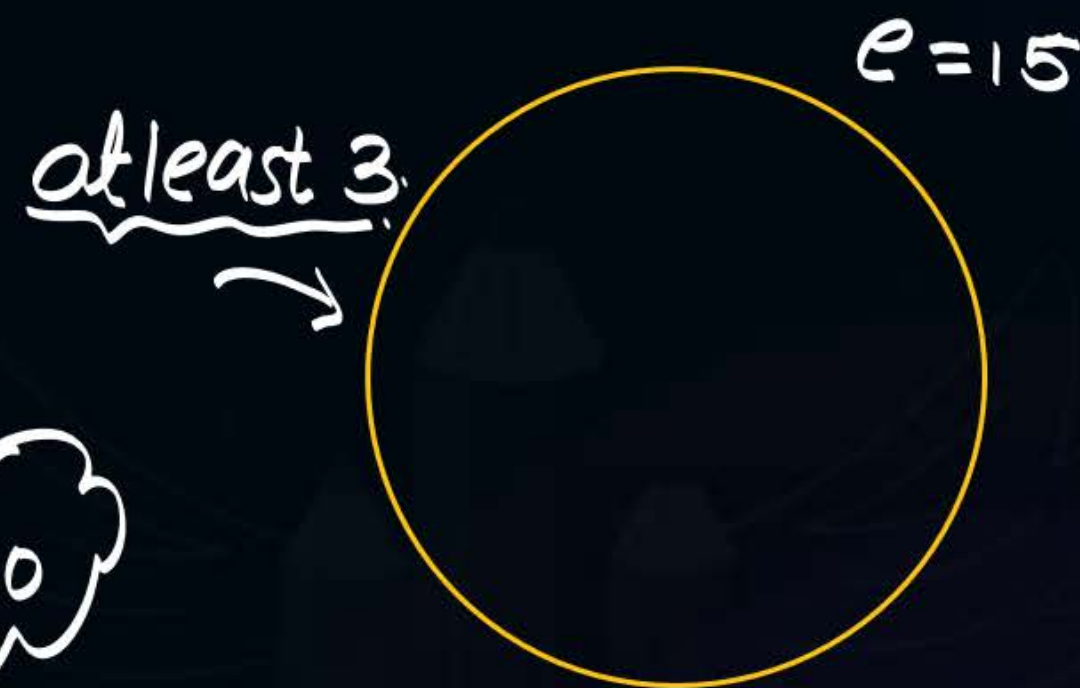
$$\delta(G) \leq \frac{2e}{n}$$

$$3 \leq \frac{2 \cdot 15}{n}$$

$$n \leq \frac{30}{3}$$

$$n \leq 10$$

Ans: 10





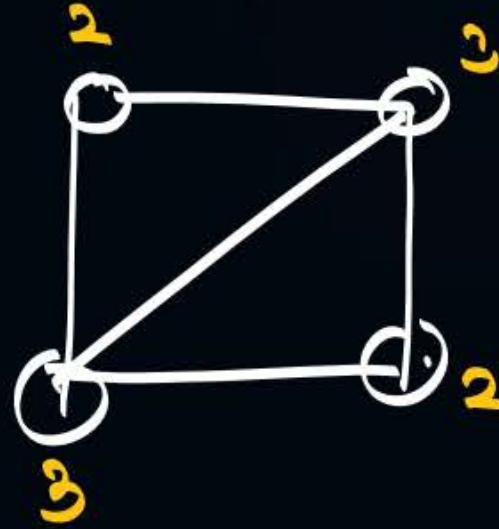
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minimum degree. ($\delta(G)$)
maximum degree ($\Delta(G)$)

Case II.

$$\delta(G) < \frac{2e}{n} < \Delta(G)$$



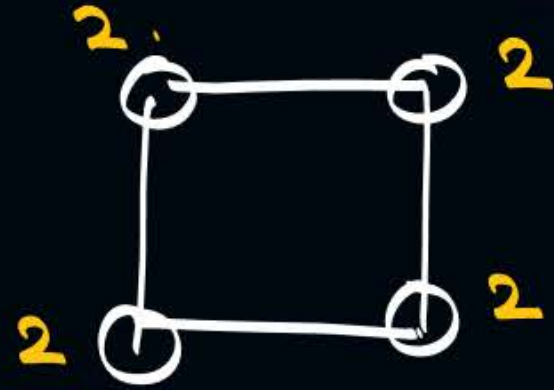
$$\delta(G) = 2$$

$$\Delta(G) = 3$$

$$\frac{2e}{n} = \frac{3+3+2+2}{4} = 2.5$$



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$$\frac{\sum d(v_i)}{n} = \frac{2e}{n} = \frac{2+2+2+2}{4} = 2.$$

case I :

$$\delta(G) = \frac{2e}{n} = \Delta(G)$$

$$\delta(G) = 2 \quad \Delta(G) = 2.$$

avg. degrees: $\frac{\text{Total degrees}}{\text{Total vertices}} = \frac{\sum d(v_i)}{n} = \frac{2e}{n}$



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Thm 5 :

$$\delta(G) \leq \frac{2e}{n} \leq \Delta(G)$$

at least

at most

Thm 3 : maximum degree
in simple graph $\leq n-1$

$$\Delta(G) \leq n-1.$$

Thm 3 & 5 :

$$\delta(G) \leq \frac{2e}{n} \leq \Delta(G) \leq n-1.$$



2 mins Summary

Topic

One

Graph Part-01

Topic

Two

Topic

Three

Topic

Four

Topic

Five

THANK - YOU